

Use Case Manual Digital Travel Credential



A Use Case Manual is a description of a specific use case for the EU Digital Identity Wallet. It includes an introduction to the use case and the user journey, the state of play of the use case development, as well as references to the relevant technical specifications and infrastructures, legislation, and governance structures.

The Use Case Manual is designed for the Wallet implementers, Member States and other stakeholders interested in the development of the Wallet and its use cases.

The Use Case Manual is not intended to contain full detailed technical specifications for the use case. However, where such specifications are available, the Use Case Manual contains pointers to them.

Description

A Digital Travel Credential (DTC) is a digital representation of the user's travel document such as a passport, identity card, or another travel document. Travel documents with chips are already composed of both a virtual and a physical component. A DTC (i.e. the virtual component as specified by ICAO) contains essentially the same information as that in the chip of a travel document*.

A EUDI Wallet user can store a DTC in their EUDI Wallet and use it for the submission of the travel document information to border authorities prior to travel, facilitating pre-screening and thereby shortening the checks performed at the border. In addition to the DTC, the user may also need to submit other travel data, such as the date of travel, flight number, etc. The reception of the DTC and other travel data is handled by a Traveller Router, which collects and verifies the information and forwards it to the relevant border authorities.

The use case applies to extra-EU travel as well as to travel to the Schengen area from Ireland and Cyprus and back. It applies to both inbound and outbound travel. Regarding the outbound travel to third countries, the goal is to facilitate pre-departure checks. Submission of DTCs to border authorities of third countries is currently out of scope of the use case and is subject to extra-EU legislation.

Technically, the submission of a DTC to entities such as airlines, hotels, embassies, banks can also be implemented, however this is not in scope of the currently proposed legislation.

User Journey(s)

Scenario 1 – DTC Issuance:

This part describes how a traveller obtains a Digital Travel Credential (DTC) and stores it in their EUDI Wallet before use.

1. Start and choice of issuance option: The user initiates the process by requesting the issuance of a DTC. Two methods are supported: scanning a physical travel document or requesting issuance directly from the competent authority.
2. Option A – Scan physical document: The user scans their physical passport or identity card using functionalities of the EU Digital Travel app. The app reads and verifies the relevant data groups and issues a DTC. In this case, eu-LISA acts as the issuing entity through the EU Digital Travel app.
3. Option B – Request from issuing authority: Alternatively, the user may request the DTC directly from the Member State's issuing authority (e.g., passport or police authority). The authority issues the DTC.

*Exceptions apply, e.g. fingerprints which are contained in eMRTDs issued by Member States, are not included in the DTC.

- 4. Store DTC in EUDI Wallet: Once issued, the DTC is securely stored in the user’s EUDI Wallet, ready to be used for pre-travel submissions and border control checks.

Scenario 2 – DTC Use:

This part outlines how a traveller uses the DTC when travelling.

- 1. Travel planning: The user plans a trip that involves crossing the external Schengen border and prepares to submit their DTC before departure.
- 2. Navigate to Traveller Router: The user accesses the Traveller Router via their EUDI Wallet. The Traveller Router serves as a secure intermediary, ensuring data is sent only to the relevant border authorities.
- 3. Submit DTC via EUDI Wallet: Using their EUDI Wallet, the traveller submits their DTC along with other required travel data (e.g., flight number, travel date).
- 4. Traveller Router receives and forwards DTC: The Traveller Router forwards it to the relevant border authority for pre-screening.
- 5. Border Authority performs Pre-Checks: The border authority receives the DTC and conducts preliminary checks before the traveller arrives.
- 6. Arrival at the border: Final checks (identity verification, travel document validation, and entry clearance) are completed using both digital and physical credentials where necessary.

Use case impact

The DTC use case enhances the travel experience by enabling seamless, secure, and privacy-preserving identity verification before and during border crossing. It generates value for travellers, service providers, and public authorities through greater efficiency, reduced processing times, and strengthened trust in travel processes.

Benefits

End-user/citizen value

Travellers benefit from faster border checks, simplified pre-travel procedures, and enhanced control over their personal data, ensuring a smoother and more secure travel experience.

- *Voluntary digital option improves convenience:*
The traveller may experience shorter waiting times at border-crossing points because pre-submission allows authorities to pre-screen. One estimate indicates ~[20 seconds average](#) saving per EU national traveller.
- *The DTC supports flexibility:*

The traveller can choose digital route or fallback to physical document, there is no mandatory shift.

- *Enhanced trust:*
By enabling verification of document authenticity/integrity ahead of physical check, the traveller benefits from smoother process, fewer ad-hoc hold-ups due to manual document checks.

Value for service providers

Airlines, airports, and other travel operators gain from automated and interoperable passenger processing systems that lower verification costs, reduce manual data handling, and improve overall operational efficiency.

- *Airlines, airports, other border crossing point operators, travel industry stakeholders benefit indirectly:*
smoother flow of travellers can reduce boarding delays, improve schedule adherence and enhance customer experience as well as improve the quality of data that is required by EU law (e.g. Advance Passenger Information (API) data by airlines) to be captured and submitted to national authorities.

Value for public authorities

Border and immigration authorities can achieve faster clearance processes, improved risk analysis, and stronger identity assurance, leading to better resource allocation and higher border security standards.

- *Border authorities:*
Advance submission of DTC enables pre-screening and risk assessment ahead of physical crossing, enabling better resource allocation, focusing on higher-risk persons rather than routine checks.
- *Member States:*
Reduced border-crossing bottlenecks, improved external border management efficiency, better alignment with digitalisation agenda and international standardisation.
- *Travel document issuers:*
Digital issuance capability opens modern service offering, can reduce reliance solely on physical document systems, improve user experience.

Economic value and potential market size

The economic value of the DTC Use Case is driven by its potential adoption across the EU’s vast travel ecosystem, encompassing millions of annual cross-border travellers, multiple national issuing authorities, and high-frequency verification transactions at airports and borders.

Users

The potential user base includes all persons who travel across the external EU borders, representing hundreds of millions of individuals.

Issuers

Issuance of travel documents, including passports and identity cards, falls under the responsibility of Member State authorities (or, in the cases of third-country nationals, authorities of third countries).

The proposed regulation on the EU Digital Travel application would, if adopted, mandate that passports and travel documents issued are accompanied by DTCs upon the applicant’s request.

DTCs can be issued to EUDI Wallets by eu-LISA using the EU Digital Travel app or by Member States using

the issuance services established for this purpose.

Relying Parties

In the context of travel, relying parties would be border authorities of Member States, as well as Iceland, Liechtenstein, Norway and Switzerland. Representatives of the border authorities would be at all international airports, seaports, train stations and road borders. As a system supporting border crossing, DTCs do not only apply to air travel but could be extended to terrestrial or maritime transport.

Transactions

According to statistics provided by Frontex, a total of [593 million crossings](#) were recorded in 2023. From these crossings, 65% were at air borders, 31% at land borders and the remaining 4% via sea borders. According to the [Eurostat monthly statistics](#), in the first 3 months of 2025 there were 208 million air passengers (all types) and the international extra-EU transport accounted for 50.6% of all passengers carried at EU level.

Synergies with other use cases

Further benefits from the use of the wallet for DTCs can be achieved through synergy with the following use cases:

- *Age verification:*
Before the availability of the Wallets, data from the chip of a travel document could be read by an age verification app (a “mini-wallet”) and used to generate a proof of age attestation. The attestation could then be used for access to age-restricted content in online platforms.
- *Identification in proximity and online scenarios in which the use of a photo is required:*
A DTC can be used as a source of an authentic photo linked to other identity attributes, to facilitate identity checks in such scenarios. (Note however that in many identification scenarios there is no need to use a photo as part of the identification process, as the wallet enables strong user authentication without it.)

Use case implementation

Existing implementations

Ongoing pilots under EUDI Wallet Large-Scale Pilots are testing DTC functionalities. The EWC pilot has been concluded, while others like [WeBuild](#) and [Aptitude](#) are building upon its findings, focusing on travel-related use cases.

Legislation

The proposed DTC framework is based on two proposals:

- [Proposal for a Regulation of the European Parliament and of the Council](#) establishing the EU Digital

Travel application (COM/2024/670 final).

- [Proposal for a Council Regulation](#) on the issuance and technical standards for digital travel credentials based on identity cards (COM/2024/671 final).

The proposals refer to ICAO specifications as a reference standard. DTCs would be in a format that enables their storage in EUDI Wallets. The EU Digital Travel app is foreseen to become operational at the end of 2030.

Use of DTCs is voluntary for travellers, but Member States would be required to accept DTCs once the legislation is adopted, and the common EU technical solution is available.

Technical specifications, software libraries and infrastructure

Data model, format and proof mechanisms

The DTC contains data equivalent to that in the electronic Machine-Readable Travel Document (eMRTD) chip. It uses ICAO 9303 and ISO/IEC TS 23220-2:2024 standards for data structure and cryptographic proofs. Verification relies partly on public key infrastructure (PKI) for the DTC itself and partly on the EUDI Wallet trust framework.

Issuance

DTCs can be created in two ways:

- *Via Public Authorities:*
To enable the issuance of DTCs, passport and identity card issuers should implement online systems whereby citizens can request and obtain a DTC based on an existing travel document issued by the same authority.
- *Via the EU Digital Travel App:*
eu-LISA should implement a functionality in the EU Digital Travel App to generate a DTC by scanning a travel document and to store it in the EUDI Wallet in the form of an electronic attestation of attributes.

Currently issued passports and identity cards with chips include information necessary for the generation of DTCs. Some Member States have established online systems for handling passport and identity card applications. These systems could potentially be extended with the issuance of DTCs to the EUDI wallet.

Verification

Verification is performed by border authorities through secure interfaces that validate the DTC’s digital signature and authenticity against passport and identity card related trusted lists (PKI). The Traveller Router forwards the DTCs to the border authorities, facilitating pre-screening while maintaining traveller privacy.

Border authorities, airports and other relevant entities should implement changes to their border control systems to enable the submission of DTCs for pre-checks. This has been piloted in [FI](#), [HR](#), [NL](#).

Further pilot projects are being planned.

